

Kali Schuchhardt

850-598-8546 | kalischuchhardt@ufl.edu | [linkedin.com/in/kalischuchhardt984](https://www.linkedin.com/in/kalischuchhardt984) | [kalischuchhardt.com](https://www.kalischuchhardt.com)

EDUCATION

University of Florida

December 2026

Bachelor of Science in Computer Engineering | Minor in EALL (Japanese) | 3.26/4.0

Gainesville, FL

Relevant Coursework: Microprocessor Applications, Reconfigurable Computing, Digital Logic, Digital Design, Signals and Systems, Physics of Electrical Engineering, Operating Systems, Circuits

Dean's List: Spring 2023, Fall 2024, Spring 2025, Fall 2025, Spring 2026

EXPERIENCE

Corrosion Engineer (Co-Op)

May 2026 – July 2026

Colonial Pipeline

Alpharetta, GA

- Developed an **Alternating Current (AC) monitoring program** by analyzing **5 years of pipeline data** to isolate high-risk locations (> 5 VAC) and optimize **Remote Monitoring Unit (RMU)** deployment
- Standardized alarm thresholds for **1,716 rectifier RMUs** by leveraging historical current density and output voltage trends to establish **statistically constrained operating alarm limits**
- Designed and executed **field tests** for **temporary Cathodic Protection coupons**, benchmarking polarization decay metrics against permanent fixtures to verify probe reliability

PROJECTS

HelpingHand – ASL Learning Smart Glove | C, ESP32, Flutter, BLE, ML

February 2026 - Present

- Designing and prototyping a wearable embedded smart glove system with flex sensors (**ADC**) and a 6-axis IMU (**I2C**) to accurately capture handshape, movement, and orientation for real-time ASL gesture recognition
- Developing robust low-latency **BLE** communication on ESP32 (**NimBLE stack**) to stream continuous sensor data at **2–5 kbps** with stable **GATT** connections to a mobile client application
- Implementing real-time haptic feedback using **PWM**-controlled vibration motors to provide precise, directional guidance for finger positioning and significantly improve user accuracy through tactile correction
- Designing a neural network-based classification pipeline to recognize ASL alphabet and numbers (**90% accuracy**)

MyFlowFriend – IoT Health Tracker | C, ESP32, Firebase, Gemini API

February 2026

- 1st Place at WiNGHacks** for an IoT digital pet tracker using an ESP32 and 3.5" TFT display
- Engineered firmware in C to drive high-speed SPI communication; implemented internal RAM **double-buffering** via **TFT_eSprite** to eliminate flickering and render smooth UI animations
- Developed an asynchronous **HTTPS POST** system to sync real-time symptom data with a **Firebase** backend, managing Wi-Fi state-logic to maintain stability on restricted 2.4GHz enterprise networks
- Integrated **Gemini 2.5 Flash API** to provide low-latency menstrual cycle predictions and an interactive chatbot based on structured daily telemetry and user symptom logs

SKILLS

Programming: Python, C/C++, Java, VHDL, SystemVerilog, ARM Assembly, AVR Assembly, MATLAB

Tools & Platforms: Linux, Certified SolidWorks Associate, Intel Quartus Prime, Waveforms, KiCAD, LTSpice

Digital & Embedded Systems: Microcontrollers and FPGAs, JTAG Debugging and Programming, SystemVerilog Coverage and Verification, FSMs, ADC/DAC, DMA, SPI, USART, PCB Design and Soldering

Languages: Japanese (JLPT N3)

LEADERSHIP

Webmaster

May 2026 – Present

Phi Sigma Rho - Sorority for Women in Engineering

Gainesville, FL

- Directing a development team to engineer a deployed **MERN** stack platform and mobile app for **80 members**
- Overseeing feature development including automated notifications, event tracking, and intuitive user interfaces
- Managing project timelines and implementing user authentication and administrative controls for role-based access

Vice President of Social Affairs

May 2025 – May 2026

Phi Sigma Rho - Sorority for Women in Engineering

Gainesville, FL

- Oversaw team of **6 event chairs** to execute of **20 major** chapter events and event logistics
- Maintained attendance records, held office hours, and coordinated with leadership on risk management